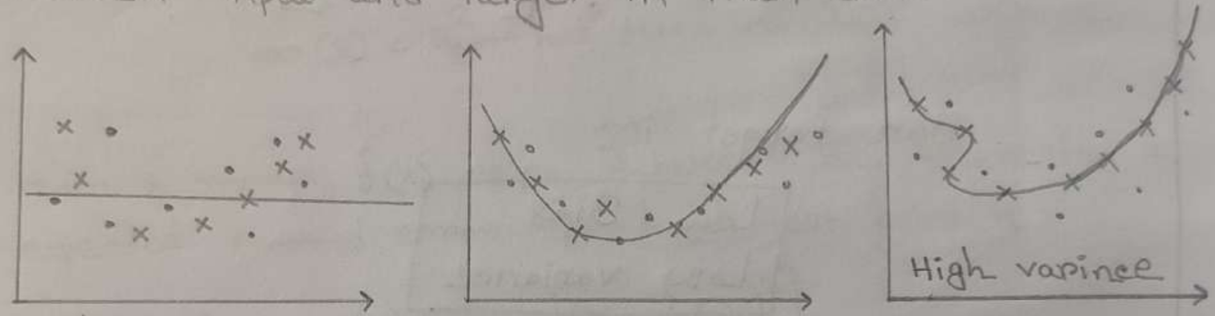
Bias Variance Trade-off:

Bias:

→ one type of error that occurs due to wrong assumptions about data.

→ Inability of a model to capture the relationship between input and target in train data.



High Bias

Low Bias

more assumptions are taken to make target function.

model fails to match the train data.

few assumptions are taken to build the target function.

. Model closely match the training data

$$Bias(\hat{y}) = E(\hat{y}) - y$$

True value

expected value of the estimator $\hat{y}$.

ways to reduce high bias:

(i) use a more complex model

(ii) increase the numbers of feature.

(iii) reduce regularization of the model

(iv) Increase the size of the training data.

variance:

→ Difference of error in training and test data.

→ amount by which the performance of a model changes when it is trained on different subsets of the training data.

$$\text{variance} = E\left[\left(\hat{y} - E(\hat{y})\right)^2\right]$$

our target is:

$\boxed{\begin{array}{l}\text{Low Bias} \\ \text{Low Variance}\end{array}}$

Low variance: less sensitive to different subsets of training data. Make consisting result in changes of training subset. Indicates under fitting

High variance: very sensitive to different subsets of training data. Changes in training subset make significant change in prediction value. [over-fitting]

Bias variance Decomposition of the MSE:

$\boxed{\text{Step-1}}$ Let's assume data generated by a function

$$Y = f(x) + \varepsilon$$

where
$f(x) \to$ true underlying function (unknown)

$\varepsilon \to$ is the noise term

$E[\varepsilon] = 0 \to$ noise has zero mean (no systematic bias)

$var(\varepsilon) = \sigma_\varepsilon^2 \to$ has fixed constant variance.

we train a model $\hat{f}(x)$ using dataset D and interested in the expected squared error at fixed test point x

$$\text{Expected Error}(x) = E_{D,\varepsilon}\left[\left(Y - \hat{f}(x)\right)^2\right]$$

$\boxed{\text{Step-2}}$ Expand the squared error $[Y = f(x) + \varepsilon]$

$$E.E(x) = E_{D\varepsilon}\left[\left(f(x) + \varepsilon - \hat{f}(x)\right)^2\right]$$

$$= E_{D\varepsilon}\left[\left\{f(x) - \hat{f}(x)\right\} + \varepsilon\right]^2$$

$$= E_{D\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)^2 + 2\left(f(x) - \hat{f}(x)\right)\varepsilon + \varepsilon^2\right]$$

$$= E_{D\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)^2\right] + E_\varepsilon\left[\varepsilon^2\right] + 2E_{D\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)\varepsilon\right]$$

Step-3

Term-1: $E_D\left[\{f(x) - \hat{f}(x)\}^2\right]$

$$E\left[(z-c)^2\right] = Var(z) + \left(E[z] - c\right)^2$$

let $z = \hat{f}(x)$

$c = f(x)$

$\mu = E_D\left[\hat{f}(x)\right]$

Then, $E_D\left[\{f(x) - \hat{f}(x)\}^2\right]$

$$= \underbrace{E_D\left[\left(\hat{f}(x) - \mu\right)^2\right]}_{\text{Variance}} + \underbrace{\left(\mu - f(x)\right)^2}_{\text{Bias}^2}.$$

$$Var_D\left(\hat{f}(x)\right) + Bias^2\left(\hat{f}(x)\right)$$

Term-2: $E_\varepsilon\left[\varepsilon^2\right]$ since $\varepsilon \sim$ noise with zero mean and σ^2 varianc

so this term is irreducible.

Term-3: $2\,E_{D,\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)\varepsilon\right]$

here $f(x)$ is fixed

$\hat{f}(x)$ is depending on D

ε is independent.

$$= 2\,E\left[f(x) - \hat{f}(x)\right] . E(\varepsilon)$$

$$= 0 \quad \left[E(\varepsilon) = 0 \text{ Bias has zero mean}\right]$$

Step-4 Final equation:

$$E_{D\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)^2\right] + E_\varepsilon\left[\varepsilon^2\right] + 2\,E_{D\varepsilon}\left[\left(f(x) - \hat{f}(x)\right)\varepsilon\right]$$

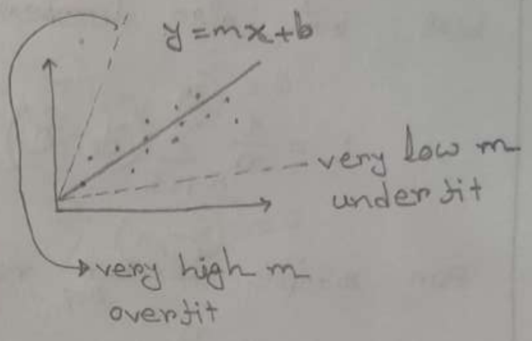
$$= \underbrace{Var\left(\hat{f}(x)\right)}_{\text{model instability}} + \underbrace{Bias^2\left(\hat{f}(x)\right)}_{\text{model assumption error}} + \sigma_\varepsilon^2 \longrightarrow \text{irreducible error (noise)}$$

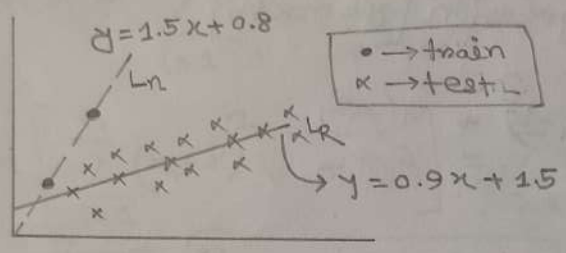
Ridge Regression:

known as L2 Regularization.

In the case of overfitting (High Variance) value of weights/ cotticients are very high.

cotticient value ↑ overfit.
cotticient value ↓ underfit.





$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \longrightarrow \text{normal mse function}$$

Although Ln will more pefect for train data and LR will make more error on Train data but it is the best fit line. so with the help of regularization we have to convence our model to choose LR instead of LN although LN is perfect on train data.

So to do this we add a penalty term to the loss function.

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda(m^2)$$

$\lambda \to$ a hyper parameter (value can be 0 to ∞)

$m \to$ slope

here we are doing square. so it's called L2 norm when we do linear regression sometimes model gets overfitted (value of cofficient and intercept becomes very high) high variance. So what we do is we added a penalty term which increase bias but also decrease the variance.

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i) + \lambda(\text{coffients}^2)$$

For simple linear regression $[y = mx + b]$

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - mx_i - b)^2 + \lambda(m^2)$$

$$\frac{\partial L}{\partial m} = 0$$

$$\Rightarrow \frac{-2}{n} \sum_{i=1}^{n} (y_i - mx_i - b).x_i + 2\lambda m = 0 \longrightarrow (1)$$

$$\frac{\partial L}{\partial b} = 0$$

$$\Rightarrow -\frac{2}{n} \sum_{i=1}^{n} (y_i - mx_i - b) = 0$$

$$\Rightarrow \sum_{i=1}^{n} \frac{y_i}{n} - m \sum_{i=1}^{n} \frac{x_i}{n} - \frac{b}{n} \sum_{i=1}^{n} 1 = 0$$

$$\Rightarrow \bar{y} - m\bar{x} - b = 0$$

$$\Rightarrow \boxed{b = \bar{y} - m\bar{x}}$$

$$L(m) = \frac{1}{n} \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$\frac{\partial L}{\partial m} = \frac{1}{n} \times 2 \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x}).(\bar{x} - x_i) + 2\lambda m = 0$$

$$\Rightarrow \frac{-2}{n} \sum (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) + 2\lambda m = 0$$

$$\Rightarrow \lambda m - \frac{1}{n}\sum_{i=1}^{n}\left[(y_i-\bar{y})-m(x_i-\bar{x})\right](x_i-\bar{x}) = 0$$

$$\Rightarrow \lambda m - \frac{1}{n}\sum_{i=1}^{n}\left[(x_i-\bar{x})(y_i-\bar{y}) - m(x_i-\bar{x})^2\right] = 0$$

$$\Rightarrow \lambda m - \frac{1}{n}\sum_{i=1}^{n}\left[(x_i-\bar{x})(y_i-\bar{y})\right] + m\cdot\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 = 0$$

$$\Rightarrow mn\lambda - \sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}) + m\sum_{i=1}^{n}(x_i-\bar{x})^2 = 0$$

$$\Rightarrow m\left[\sum_{i=1}^{n}(x_i-\bar{x})^2 + n\lambda\right] = \sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

$$\Rightarrow \boxed{m = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{n\lambda + \sum_{i=1}^{n}(x_i-\bar{x})^2}}$$

$\lambda = 0 \longrightarrow$ Linear Regression

For n-dimensional data

given a dataset

$X \in R^{m\times n} \longrightarrow$ m rows and n featurs(columns)

$y \in R^{m\times 1} \longrightarrow$ target column(vector)

we want to find weights $W \in R^{n\times 1}$ and intercept $b \in R$

$\hat{y} = XW + b.1_m$ $\quad$ $1_m \in R^m$ so $b.1_m \rightarrow$ a column vector broadcasting

$\boxed{\text{step-1}}$ Ridge Regression Cost Function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left\{y_i + (x_i.w+b)\right\}^2 + \underbrace{\frac{\alpha}{2m}\|w\|_2^2}_{\text{L2 norm/Shrinkage coef}}$$

here $\|w\|_2^2 = w_1^2 + w_2^2 + \cdots + w_n^2$

$\alpha \geq 0 \longrightarrow$ strength of regularization

$\boxed{\text{step-2}}$

$\hat{y} = XW + b1$ $\quad$ here $X \in R^{m\times n}$ $\quad$ $1 \in R^{m\times 1} \longrightarrow$ a column matrix of one

$W \in R^{n\times 1}$

$$J(w,b) = \frac{1}{2m}\|y - xw - b1\|^2 + \frac{\alpha}{2m}\|w\|^2$$

$\boxed{\text{step-3}}$ Find gradients

* gradient w.r.t W:

$$\frac{\partial}{\partial W}\left(\frac{1}{2m}\|y-(XW+b)\|^2\right) \longrightarrow \underline{\text{Term-1}}$$

$\Rightarrow$ let $e = y-(XW+b)$ then

$$= \frac{\partial}{\partial W}\left(\frac{1}{2m}\|e\|^2\right) \Rightarrow$$

$$= \frac{\partial}{\partial W}\left(\frac{1}{2m} e^T e\right) \quad \left[\|e\|^2 = e^T e\right]$$

$$= \frac{\partial}{\partial W}\left(\frac{1}{2m} (y-XW-b)^T (y-XW-b)\right)$$

$$= \frac{-1}{2m} 2X^T (y-XW-b)$$

$$= -\frac{1}{m} X^T (y-XW-b) \longrightarrow \text{gradient of error term}$$

now $\frac{\partial}{\partial W}\left(\frac{\alpha}{2m}\|W\|^2\right) =$

$$= \frac{\partial}{\partial W}\left(\frac{\alpha}{2m} \cdot W^T W\right)$$

$$= \frac{\alpha}{2m} \times 2W = \frac{\alpha}{m} \cdot W$$

$$\boxed{\frac{\partial J}{\partial W} = -\frac{1}{m} X^T (y-XW-b) + \frac{\alpha}{m} W}$$

gradient w.r.t b

$$\frac{\partial J}{\partial b} = \frac{\partial}{\partial b}\left(\frac{1}{2m}\|y-(XW+b)\|^2\right)$$

$$= \frac{\partial}{\partial b}\left(\frac{1}{2m}\sum_{i=1}^{m}(y_i - W^T x_i - b)^2\right)$$

$$= -\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)$$

$W \in R^{n \times 1}$, $x_i \in R^{n \times 1}$

But we want product of x and W

$W^T x_i \in R \rightarrow$ a scalar

$y_i \rightarrow$ also a scalar

$$= -\frac{1}{m} 1^T (y - XW - b)$$

$\boxed{\text{Note}}$ multiply a vector matrix with a Identity matrix doesn't change the vector $\boxed{A.I = A}$

$\therefore$ Final grandients

$$\frac{\partial J}{\partial w} = -\frac{1}{m} X^T (y - \hat{y}) + \frac{\alpha}{m} \cdot w$$

and

$$\frac{\partial J}{\partial b} = -\frac{1}{m} \sum^{m} (y - \hat{y})$$

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using OLS

Goal: $J(w) = \|y - Xw\|^2 + \alpha \|w\|^2$

$X \in R^{m \times n} \longrightarrow$ input matrix

$y \in R^m \longrightarrow$ target vector

$w \in R^n \longrightarrow$ weight vector

$\alpha \longrightarrow$ regularization Strength

$\boxed{\text{Step-1}}$ expand the loss function

$$J(w) = (y - Xw)^T (y - Xw) + \alpha w^T w \quad \left[\sum A^2 = A^T A\right]$$

$$= (y^T - w^T X^T)(y - Xw) + \alpha w^T w \quad \left[(AB)^T = B^T A^T ,\ (A-B)^T = A^T - B^T\right]$$

$$= y^T y - y^T X w - w^T X^T y + w^T X^T X w + \alpha w^T w$$

$$= y^T y - 2 y^T X w + w^T X^T X w + \alpha w^T w \quad \left[\begin{array}{c} y^T X w \text{ and } w^T X^T y \\ \text{is symetric} \end{array}\right]$$

$a^T w$ or $w^T a$ then

$\nabla_w (a^T w) = \nabla_w (w^T a) = a$

$\nabla_w (w^T A w) = 2Aw$

if A is symmetric

$\boxed{\text{Step-2}}$ Take part by part derivative

$\frac{\partial}{\partial w}(y^T y) = 0$

$\frac{\partial}{\partial w}(-2y^T x w) \longrightarrow$ take it like this $\frac{\partial}{\partial w}\left(-2a^T w\right)$ where $a^T = x^T y$

$\nabla_w (-2y^T x w) = -2X^T y$

why we use this flip?

$y^T x w = w^T x^T y =$ scalar

cause we can't do derivative of w just. so must form like $w^T A w \rightarrow 2Aw$ or

$\nabla_w (a^T w) = a$ or $\nabla_w (w^T a) = a$

so instead of writting

$y^T x w$ we write $w^T x^T y$

scalar

$(\text{scalar})^T = \text{scalar}$

$(y^T x w)^T = w^T x^T y$

set gradient $= 0$

$J(w) = -2x^T y + 2x^T x w + 2\alpha w = 0$

$\Rightarrow -x^T y + x^T x w + \alpha w = 0$

$\Rightarrow w(x^T x + \alpha I) = x^T y$

$\Rightarrow \boxed{w = (x^T x + \alpha I)^{-1} x^T y}$ $\longrightarrow$ give solver='cholesky' in Ridge to use this

"Lasso Regression"

Lasso add an extra penalty term to the cost function. By doing this cost function shrinks less relavant feature's cofficients to zero.

In this way it's remove the less relavant features.

The cost function for LASSO:

$$J = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{m}|w_j|$$

Why Lasso Regression Creates Sparsity?

→ Sparsity: when we increase the value of λ at a very high point some of the features coefficents becomes zero.

For single input feature

Linear	Ridge	Lasso
$m = \dfrac{\sum_{i=1}^{n}(y_i - \bar{y}_i)(x_i - \bar{x}_i)}{\sum_{i=1}^{n}(x_i - \bar{x}_i)^2}$	$m = \dfrac{\sum_{i=1}^{n}(y_i - \hat{y}_i)(x_i - \bar{x}_i)}{\sum_{i=1}^{n}(x_i - \bar{x}_i)^2 + \lambda}$	$m = \dfrac{\quad}{\sum}$

Lasso loss function

$$L = \sum_{i=1}^{n}(y_i - mx_i - b)^2 + \lambda|m|$$

$$|m| = \begin{cases} m \; ; & m > 0 \\ -m \; ; & m < 0 \end{cases}$$

$$|m| \rightarrow \begin{cases} m & ; \; m > 0 \\ 0 & \; m = 0 \\ -m & \; m < 0 \end{cases}$$

when $m > 0$

$$L = \sum_{i=1}^{n} (y_i - mx_i - b)^2 + 2\lambda m$$

$$L = \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})^2 + 2\lambda m \qquad \left[b = \bar{y} - m\bar{x} \right] \text{ note}$$

$$\frac{\partial L}{\partial m} = 2 \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda = 0$$

$$\Rightarrow -2 \sum \left\{ (y_i - \bar{y}) - m(x_i - \bar{x}) \right\} (x_i - \bar{x}) + 2\lambda = 0$$

$$\Rightarrow - \sum (y_i - \bar{y})(x_i - \bar{x}) + \lambda + m \sum (x_i - \bar{x})^2 = 0$$

$$\Rightarrow \quad m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum (x_i - \bar{x})^2}$$

Lasso

$\boxed{m > 0}$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum (x_i - \bar{x})^2}$$

$\boxed{m < 0}$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum (x_i - \bar{x})^2}$$

$\boxed{m = 0} \rightarrow$ Linear Reg.

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

this is >0 cause m>0

$$m > 0 \rightarrow m = \frac{\sum(y_i-\bar{y})(x_i-\bar{x}) - \lambda}{\sum(x_i-\bar{x})^2}$$

as $\lambda \uparrow$ $\left[\sum(y_i-\bar{y})(x_i-\bar{x}) - \lambda\right] \downarrow$ at a high value of λ

$\sum(y_i-\bar{y})(x_i-\bar{x}) == \lambda \longrightarrow m=0$ (sparsity)

$$m < 0 \rightarrow m = \frac{\sum(y_i-\bar{y})(x_i-\bar{x}) + \lambda}{\sum(x_i-\bar{x})^2}$$

$m<0$ so $\sum(y_i-\bar{y})(x_i-\bar{x}) < 0$

as $\lambda \uparrow \Rightarrow \left[\sum(y_i-\bar{y})(x_i-\bar{x}) + \lambda\right]$ moves towards zero

and when $\lambda == \left|\sum(y_i-\bar{y})(x_i-\bar{x})\right|$ at this point $m=0$

That's why Lasso Regression with very high λ makes the coefficient matrix to zero.

Ridge → when all features are important

Lasso → Subset of collumns are important (Feature Selection)

Large dataset or multicolinearity ← ElasticNet → High numbers of collumns and it's not possible to gather the knowledge about all collumns

Loss function: $\boxed{L = \sum(y_i-\hat{y}_i)^2 + a\|m\|^2 + b\|m\|}$

$\lambda = a+b$ $\quad l_1 = \frac{a}{a+b}$

Scikit-Learn hypermeter

$\lambda = 10$ and l_1 ratio $= 0.9$ mean use 90% ridge and 10% Lasso with $\lambda = 10$